ME 333 - Mechanics of Materials

Syllabus & Course Proceedings — (Aug - Nov 2025)

Lecture Information:

Course Title	:	Mechanics of Materials
Instructor & Tutor	:	Prof. Ravi Sastri Ayyagari
Laboratory Instructor	:	Prof. Harmeet Singh
Lecture Hours	:	10:00 am - 11:20 am [Wed & Fri - 7/109]
Tutorial Hours	:	2:00 pm - 3:20 pm [Mon - 7/109]†
Laboratory Hours	:	2:00 pm - 3:20 pm [Mon - 4/101]†
Course Material	:	Google Classroom - Class code: lluyxsvu
Email	:	ravi.ayyagari@iitgn.ac.in; harmeet.singh@iitgn.ac.in
Office	:	AB - 4/309 (Prof. Ravi)
		AB - 4/301 (Prof. Harmeet)

† The tutorial and lab sessions will be held alternately starting with the first laboratory session on 11$^{\text{th}}$ August. An introductory course outline session will be held on 4$^{\text{th}}$ August.

Pre-requisite:

ES 221 – Mechanics of Solids

Course Content:

1. Strain at a point, strain-displacement relations in Cartesian coordinates, compatibility equations - [Practice Problems 1]

2. Stress at a point, equilibrium equations, 3-D state of stress, principal stresses and maximum shear stress, stress invariants - [Practice Problems 2]

3. Stress-Strain relations, concept of yielding - [Practice Problems 3 & 4]

4. Torsion & Torsion of non-circular shafts - [Practice Problems 5]

5. Bending, unsymmetrical bending, shear flow and shear center; Symmetric bending of curved bars - [Practice Problems 6, 7 & 8]

6. Formulation of elasticity problem, boundary conditions, plane stress and plane strain problems. Airy stress function, solution by polynomials - [Practice Problems 9]

7. 2-D problems in polar cylindrical coordinates, equilibrium equations and strain-displacement relations, thick cylinders, composite tubes, Stress distribution around a circular hole, stress concentration factor - [Practice Problems 10]

Note: The order of course content may slightly vary from that outlined above depending on the pace and logical flow of the topics covered during lectures. Typically each topic on an average will be covered in about $3-4$ lectures.

Text & References

1. Arthur P. Boresi, Richard J. Schmidt, Advanced Mechanics of Materials; Sixth Edition, Wiley India Pvt. Ltd., 2002. [Prescribed Text]
2. S. P. Timoshenko, Strength of Materials Part 2: Advanced Theory and Problems; Third Edition, CBS, 2002.
3. J. R. Barber, Intermediate Mechanics of Materials; Second Edition, Springer, 2010.
4. Ansel C. Ugural, Saul K. Fenster, Advanced Mechanics of Materials and Applied Elasticity; Fifth Edition, Prentice Hall, 2011.
5. R.D. Cook and W.C. Young, Advanced Mechanics of Materials; Second Edition, Prentice Hall, 1999.
6. E. P. Popov, Engineering Mechanics of Solids; Second Edition, Pearson, 1998.

Students are strongly encouraged to prepare their own notes after attending the lectures. Course material as slides will be shared on google classroom after each lecture.

Attendance:
Attendance is not mandatory. However, students are strongly encouraged to attend all lectures.

Practice Problems:
There will be $8-10$ Practice problem sets assigned in the course. These problem sets are to enable students to gain a deeper conceptual understanding. It is not required for students to submit the worked out problems and no grade will be awarded for solving these problem sets. Nevertheless, solving these problems independently will allow you to better perform in your Quizzes, mid- and end-sem exams. Note that learning the material to complete the practice problems is the sole responsibility of the student.

Laboratory Sessions:
The laboratory sessions aid students to fully comprehend the material covered in lectures. Guidelines for laboratory report submissions and details of experiments to be carried out will be shared separately through Google Classroom.

Grading basis:
Quiz (tentatively 4) 25%, Laboratory Reports (tentatively $5-6$) 25%, Mid-Term Exam 25% and Final Exam 25%. A cumulative score of less than 40% will result in a 'F' grade.

Collaboration policy:
Students are expected that they work on the practice problems on their own. This will actually help them gain confidence on the material learnt. Also, it will allow them to calibrate the required effort for successful completion of the course. Material submitted for grading by a student should represent his/her own work. It is not acceptable to copy the work from other sources, and more so, it is not acceptable for two students to submit identical copies of any part of an assignment. Students are expected to comply with the honor code of the Institute. Plagiarism of any form in assignments and/or exams will guarantee an 'F' grade.